End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

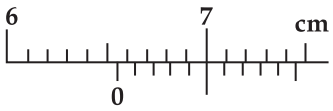


Fig (1.26)

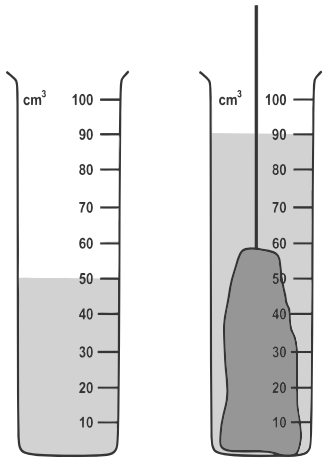


Fig (1.27)

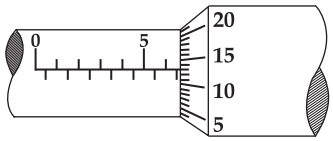


Fig (1.28)

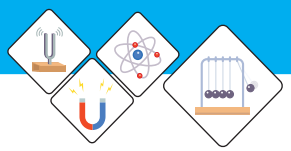
1. The Figure (1.26) shows part of a Vernier scale, what is the reading on the Vernier scale:
   a) 6.50 cm b) 6.55 cm
   c) 7.00 cm d) 7.45 cm

2. Ten identical steel balls each of mass 27g, are immersed in a measuring cylinder having 20cm^3 of water. The reading of water level rises to 50cm^3. What is the density of the steel?
   a) 0.90 gm/cm^3 b) 8.1 gm/cm^3
   c) 9.0 gm/cm^3 d) 13.5gm/cm^3

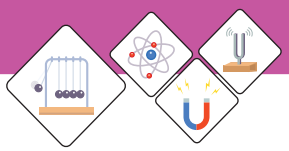
3. An object of mass 100g is immersed in water as shown in the figure (1.27), what is the density of the material from which object is made?
   a) 0.4gcm^3 b) 0.9gcm^3
   c) 1.1 gcm^3 d) 2.5gcm^3

4. What is the reading of this micrometer in figure (1.28)?
   a) 5.43mm b) 6.63mm
   c) 7.30mm d) 8.13mm

5. A chips wrapper is 4.5 cm long and 5.9 cm wide. Its area upto significant figures will be
   a) 30 cm^2 b) 28 cm^2
   c) 26.55 cm^2 d) 32 cm^2



6. A worldwide system of measurements in which the units of base quantities were introduced is called:
   a) prefixes
   b) international system of units
   c) hexadecimal system
   d) none of above
7. All accurately known digits and first doubtful digit in an expression are known as:
   a) non-significant figures
   b) significant figures
   c) estimated figures
   d) crossed figures
8. If zero line of Vernier scale coincides with zero of main scale, then zero error is:
   a) positive b) zero
   c) negative d) one
9. zero error of the instrument is:
   a) systematic error b) human error
   c) random error d) classified error
10. Length, mass, electric current, time, intensity of light and amount of substance are examples of:
   a) base quantities b) derived quantities
   c) prefixes d) fundamental quantities



End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. Scalar Quantities have ..... and suitable unit.
   a) magnitude b) direction c) both (a) and (b)
2. Vector quantities have ..... along with magnitude and unit.
   a) Magnitude b) direction c) both (a) and (b)
3. Which one is a vector quantity .....
   a) Mass b) Weight c) time
4. Which one is a scalar Quantity.....
   a) time b) Force c) Velocity
5. Distance is a .....quantity.
   a) Vector b) Scalar c) both (a) and (b)
6. What is SI unit of acceleration
   a) ms^{-1} b) ms^{-2} c) Nm
7. What is a SI Unit of Velocity.....
   a) Nm b) ms^{-1} c) ms^{-2}
8. Shortest distance between two points is called
   a) distance b) speed c) displacement

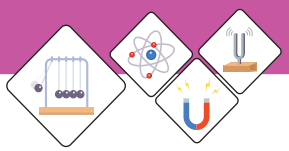
Section (B) Structured Questions

Rest and motion

a) Define rest and motion.
b) What is meant by relative motion?

Types of motion

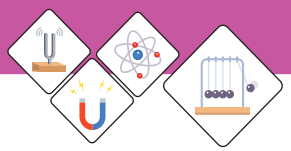
a) Define speed and velocity.
b) What is difference between distance and displacement?



End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. Newton's First law of motion is also known as law of
   a) speed b) rest
   c) inertia d) force
2. Quantity of matter contained in body is called_____.
   a) mass b) volume
   c) area d) weight
3. Quantity of motion contained in a body is called
   a) force b) inertia
   c) momentum d) gravity
4. Law of conservation of momentum defines that the total momentum of a system of two bodies before and after collision_____.
   a) remains constant
   b) retains more momentum
   c) losses some momentum
   d) None of above is true
5. Weight of a body can be measured using a spring balance, it differs from place to place because of variation in _____.
   a) acceleration b) gravitational pull
   c) velocity d) size of spring balance
6. It is easier to push an empty shopping cart than a full one, because the filled cart has more mass than the empty one. This can be expressed by
   a) $F > m$ b) $F < m$
   c) $F \alpha 1/m$ d) $F \propto m$

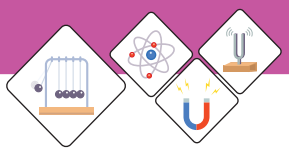


7. Centrifugal force is always directed:
   a) towards centre
   b) away from centre
   c) along the circular path
   d) all sides
8. Friction opposes motion between two bodies in contact because of:
   a) charges on bodies
   b) weight of bodies
   c) roughness of surfaces
   d) None of above
9. Which statement is true for limiting frictional force.
   a) it is greater than rolling friction
   b) it is greater that sliding friction
   c) it is greater that kinetic friction
   d) all are true
10. A man pulls a crate of mass 25 kg across leveled ground with a horizontal force of 60 N.A constant force of friction of 20 N acts on the sledge. What is the acceleration of the sledge?
   a) $0.63\ ms^{-2}$ b) $1.6\ ms^{-2}$
   c) $2.4\ ms^{-2}$ d) $3.2\ ms^{-2}$

Section (B) Structured Questions

Momentum

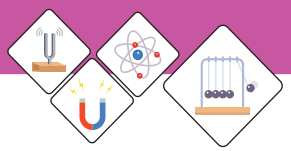
1. a) Define momentum with SI unit.
   b) Find the moment of body of mass 6 kg moving with a velocity of $25\ ms^{-1}$.
   c) What will be the velocity if the momentum becomes 200Ns?



End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. A pair of unlike parallel forces having different lines force produce----------.
   a) equilibrium b) torque
   c) a couple d) unstable equilibrium
2. Head-to-tail rule can be used to add---------- forces.
   a) two b) three
   c) five d) any number of
3. A force of 15 N makes an angle of 60° with horizontal. Its vertical component will be:
   a) 15N b) 10N
   c) 13N d) 7 N
4. A body is in equilibrium when it has
   a) uniform speed b) uniform acceleration
   c) both (a) and (b) d) zero acceleration
5. A body is in stable equilibrium after slight tilt if its Centre of gravity
   a) remains above the point of contact
   b) remains on one side of point of contact
   c) passes over the point of contact
   d) is at lowest position
6. A body is in unstable equilibrium after slight tilt if its center of gravity
   a) remains on one side of the point of contact
   b) remains above the point of contact
   c) passes over the point of contact
   d) is positioned at its bottom



7. A body is in neutral equilibrium when its Centre of gravity
   a) Is at the lowest position
   b) Remains at same height
   c) Is at highest position
   d) Is at its base
8. Bunsen burner is made stable by
   a) Increasing its length
   b) Increasing its mass
   c) Decreasing its base area
   d) Increasing its base area
9. A tight rope walker carries a long pole to
   a) Increase his weight
   b) Raise his Centre of gravity
   c) Lower his Centre of gravity
   d) Keep his Centre of gravity in fixed position
10. Stability of a racing car is increased by
   a) Increasing its height
   b) Raising its Centre of gravity
   c) Decreasing its width
   d) Lowering its Centre of gravity

End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. The springs in brakes and clutches are used
   a) To restore original position
   b) To measure forces
   c) To absorb shocks
   d) To absorb strain energy
2. If the material recovers the original dimensions, when an external force is removed, this deformation is known as _____ deformation.
   a) Inelastic b) Permanent
   c) Elastic d) Irreversible
3. Which of the following material is more elastic?
   a) Rubber b) Glass
   c) Steel d) Wood
4. If a spring stretches easily then its spring constant has _____.
   a) Large value b) Small value
   c) Constant Value d) Both (a) and (b)
5. What is the unit for the spring constant?
   a) Nm b) Nm^{-2}
   c) Nm^{-1} d) Nm^{2}
6. The spring obeys Hooke's law for the earlier extensions and when the spring becomes damaged it does not appear to do so; Fig 5.12. Estimate, from graph, after addition of which weight the spring damaged.
   a) 1.5 N b) 8 N
   c) 1.6 N d) 2.0 N
7. Which of the following is not a unit of pressure?
   a) Pascal b) Bar
   c) Atmosphere d) Newton

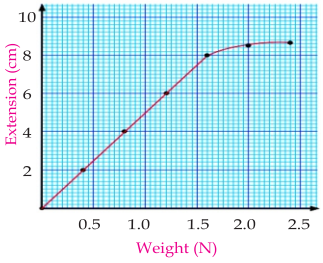


Fig (5.12)
Graph between weight and extension for spring

8. If a metal block applies a force of 20 N on an area of 5 cm^2. Find the pressure being applied by the block on the area of ____.
   a) 100 Ncm^{-2} b) 0.8 Ncm^{-2}
   c) 0.25 Ncm^{-2} d) 4 Ncm^{-2}
9. The Fig (5.13) shows a container with three spouts. The container is filled with water. Jets of water pour out of the spouts. Why does the jet of water from the bottom spout goes farthest out from the container?
   a) Pressure decreases with depth.
   b) Pressure increases with depth.
   c) More water available to flow out from the bottom.
   d) Density of water different at different places.

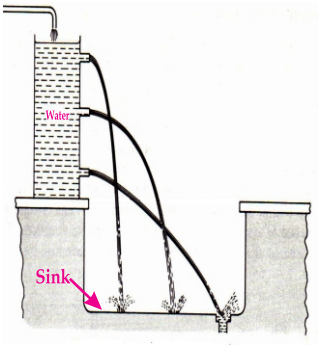


Fig (5.13)

Section (B) Structured Questions

Stretching of Spring

1. Some students experimented to find out how a spring stretched when loads were added to it

Table (5.3) For load and extension

Load (N)	0	2	4	6	8	10	12	14
Extension (mm)	0	15	30	45	60	75	90	100

   a) Use these results to plot a graph. (Plot x = load, y = extension in spring).
   b) Use your graph to find
      i) The extension when the load is 3 N;
      ii) The load which produces an extension of 40mm.
2. The variation in extension x of the force (F) for a spring is shown in Fig (5.14). The point (L) on the graph is the elastic limit of the spring.
   a) Describe the meaning of elastic limit.

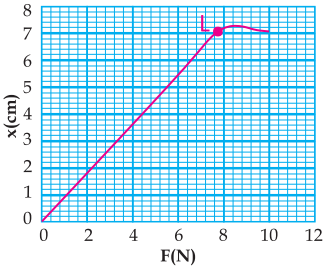
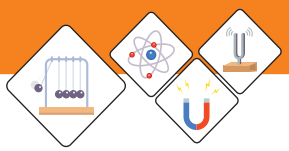


Fig (5.14)
Graph between force loadand extension



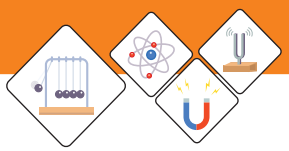
End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. The motion of a falling ball towards Earth is due to the ----------.
   a) Weightlessness
   b) Gravitational force
   c) Acceleration due to gravity
   d) Both 'a' and 'b'
2. Newton's law of gravitation holds between every two objects on the ----------.
   a) on Earth b) on Jupiter
   c) on Moon d) on Universe
3. Numerical value of G is ----------.
   a) $G = 6.673 \times 10^{-11} Nm^2 kg^{-2}$
   b) $G = 6.673 \times 10^{11} Nm^2 kg^{-2}$
   c) $G = 6.763 \times 10^{-10} Nm^2 kg^{-2}$
   d) $G = 6.763 \times 10^{11} Nm^2 kg^{-2}$
4. Gravitational field of Earth is directed ----------.
   a) towards the Earth b) towards the Sun
   c) towards the Moon d) away from Earth
5. ---------- was the first scientist who gave the concept of gravitation.
   a) Einstein b) Newton
   c) Faraday d) Maxwell
6. According to Newton's law of universal gravitation force ($\propto$) _____.
   a) $m_1 m_2$ b) $\frac{1}{r^2}$
   c) r^2 d) Both (a) and (b)
7. Gravitational force is always ----------.
   a) Repulsive b) Attractive
   c) Both a and b d) None of these



8. Numerical value of ---------- remains constant every where.
   a) g b) G
   c) F d) W
9. Gravitation force is ---------- of the medium between the objects.
   a) Dependent b) Independent
   c) Both 'a' and 'b' d) None of these
10. Near Earth's surface (g) = ----------.
   a) $10ms^{-2}$ b) $1.6ms^{-1}$
   c) Both (a) and (b) d) None of these
11. Newton's law of gravitation is consistent with Newton's ---------- law of motion.
   a) 1st b) 2nd
   c) 3rd d) All of them
12. Spring balance is used to measure ----------.
   a) Mass b) Weight
   c) Elasticity d) Density
13. Your weight as measured on Earth will be --------- on Moon.
   a) Increased b) Decreased
   c) Remains same d) None of these
14. Mass of Earth is ----------.
   a) $6.0 \times 10^{23}kg$ b) $6.0 \times 10^{24}kg$
   c) $6.0 \times 10^{25}kg$ c) $6.0 \times 10^{26}kg$
15. ---------- is a natural satellite.
   a) Earth b) Jupiter
   c) Moon d) Mars
16. A communication satellite completes its one revolution around the Earth in ---------- hours.
   a) 6 b) 12
   c) 18 d) 24
17. The velocity of a satellite is ---------- of its mass.
   a) Independent b) Dependent
   c) Equal d) Double



18. ---------- are used to put satellites into orbits.
    a) Helicopter b) Aeroplane
    c) Rocket d) None of these
19. The critical velocity (v_c) = ----------.
    a) gR b) $\frac{g}{R}$
    c) $\sqrt{gR}$ d) $\sqrt{\frac{g}{R}}$

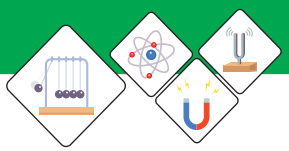
Section (B) Structured Questions

Newton's Law of Gravitation

1. a) Why we do not feel the gravitational force of attraction from the objects around us?
   b) Define Gravitational field with an example.
2. a) Write down any three characteristics of Gravitational force.
   b) Define gravitational field strength.
3. a) State and explain Newton's law of gravitation.
   b) Define 'field force'.
4. Determine the gravitational force of attraction between Urwa and Ayesha standing at a distance of 50m apart. The mass of Urwa is 60kg and that of Ayesha is 70kg.

Weight

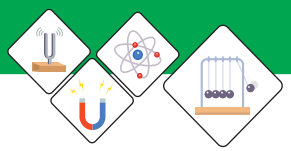
5. a) Why weight of an object does not remain same every where on Earth?
   b) Why the unit of weight is Newton? Explain.
6. a) Define weight and write down its equation.
   b) Weight of Rani is 450N at the surface of Earth. Find her mass?
7. Weight of Naveera is 700N on the Earth's surface. What will be Naveera's weight at the surface of Moon?



End of Unit Questions

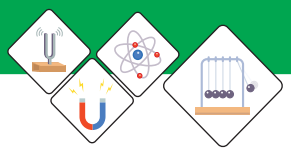
Section (A) Multiple Choice Questions (MCQs)

1. An object with particles close together and vibrating describes a_____.
   a) Gas b) Liquid
   c) Solid d) All three
2. A burning candle is an example of _____ state of matter.
   a) Gas b) Liquid
   c) Solid d) All three
3. During which process a gas becomes a liquid _____.
   a) Melting b) Freezing
   c) Condensing d) Boiling
4. A solid can _____.
   a) have a fixed shape
   b) be easily compressed
   c) take a shape of container
   d) have freely moving molecules (KMT).
5. According to kinetic molecular theory, the pressure exerted by a gas is caused by the _____.
   a) bombardment of the gas molecules on the walls of the container.
   b) collision between gas molecules.
   c) large distances between gas molecules.
   d) random motion of the gas molecules.
6. If a gas is heated in a sealed cylinder, then_____ increases.
   a) pressure inside the container
   b) average kinetic energy of the particles
   c) temperature of the gas
   d) All of them



7. A gas in a container of fixed volume is heated. What happens to the molecules of the gas?
   a) They collide less frequently.
   b) They expand.
   c) They move faster.
   d) They move further apart.
8. In a liquid, some energetic molecules break free from the surface even when the liquid is too cold for bubbles to form. What is the name of this process?
   a) boiling b) condensation
   c) convection d) evaporation
9. What happens to the molecules of a gas when the gas changes into a liquid?
   a) They move closer and lose energy.
   b) They move closer and gain energy.
   c) They move apart and lose energy.
   d) They move apart and gain energy
10. A substance has a melting point of –17 °C and a boiling point of 117 °C. In which state does the substance exist at –10 °C and at 110 °C?

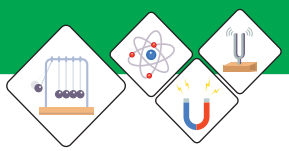
	at -10^{0}C	at 110^{0}C
a	Solid	liquid
b	solid	gas
c	liquid	liquid
d	liquid	gas



End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. If force of 6N displaces an object 2m in the direction of force, then work done will be_____.
   a) 0 Joule b) 12 Joule
   c) 3 Joule d) 12 N.
2. If a body of mass 1 kg is moving with velocity of 1m/sec then K.E of the body will be _____.
   a) 10 Joules b) 0.5 Joules
   c) 2 Joules d) 1 Joule
3. If a machine performs 20J of work in 10sec then its power is _____.
   a) 200 watt b) 20 watt
   c. 2watt d) 0.2 watt
4. A body of mass 1kg is lifted through a height of 1m . The energy possessed in the body will be_____. (consider $g = 10ms^{-2}$)
   a) 1J b) 10 Joule
   c) 100 Joule d) 1000 Joule
5. The energy released during fission or fusion reaction is called _____.
   a) Solar energy b) Geothermal energy
   c) Tidal energy d) Nuclear energy
6. Which is the renewable source of energy
   a) Solar and wind b) Coal
   c) Natural gas d) Petroleum
7. The ratio of output to input is called
   a) Energy b) Work
   c) Power d) Efficiency



8. Work done per unit time is called _____.
   a) Efficiency b) Energy
   c) Power d) Force
9. Coal, gas and oil are all examples of_____.
   a) Tidal energy b) Nuclear energy
   c) Fossil fuel energy d) Biomass energy
10. _____ is not a renewable source of energy.
   a) Solar energy b) Coal
   c) Wind energy d) Geothermal energy

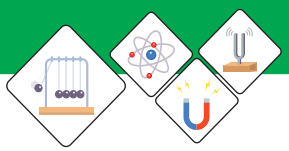
Section (B) Structured Questions

Work

1. a) Define work.
   b) Derive the equation; (Work = Fd cosθ).
2. How much work is needed to move horizontally a body 20m by a force of 30N, the angle between the body and the horizontal surface is 60°?
3. How much work is done, if a crate is moved at a distance of 50m, when a force of 30N is applied along the surface.
4. What is the work done by Usman? If a bar of weight 100 N is brought by him from (A) to (B), then brought back to (A).

Energy Forms

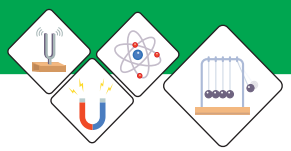
5. a) Define Kinetic energy
   b) Derive the equation (K·E = $1/2\,mv^2$).
6. What will be the Kinetic energy of a boy of mass 50kg driving a bike with velocity of $2ms^{-1}$?
7. a) Define Potential Energy.
   b) Derive the equation (P·E = mgh).



End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. Heat is the form of _____
   a) Pressure b) Weight
   c) Energy d) All
2. Heat capacity is the product of mass and _____
   a) Boiling point b) Freezing point
   c) Energy d) Specific heat of material
3. The amount of heat needed to convert a substance from liquid to gas is called _____
   a) Heat of Vaporization b) Specific heat
   c) latent heat of fusion d) All
4. Thermal energy transfer required per unit mass to increase the temperature by 1^0C or 1 K is called ____
   a) Latent heat of Vaporization
   b) Specific heat capacity
   c) Latent heat of fusion
   d) Thermal capacity
5. A fixed temperature at which a pure liquid boils is called_____.
   a) melting point b) freezing point
   c) boiling point d) Both (a) and (b).
6. The melting point of ice at normal atmospheric pressure is _____.
   a) 0^0C b) 0K
   c) 100^0C d) Both (a) and (b)



7. Thermal energy transfer required to change a solid into liquid without changing its temperature is called _____.
   a) Latent heat of fusion
   b) latent heat of vaporization
   c) latent heat of boiling
   d) specific heat capacity
8. Thermal energy transfer required to change a liquid into gas without changing its temperature is called
   a) latent heat of freezing
   b) latent heat of vaporization
   c) latent heat of boiling
   d) latent heat of melting
9. Vaporation can occur at _____
   a) freezing point b) melting point
   c) boiling point d) all temperatures
10. Rate of evaporation of a liquid can be increased by_____.
   a) increasing humidity
   b) decreasing temperature
   c) increasing its boiling point
   d) decreasing atmospheric pressure
11. Linear thermal expansion of a solid depends upon_____.
   a) change in temperature
   b) original length
   c) properties of material
   d) all of these